


THE UNIVERSITY OF
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Final Report

# The Art of Scientific Computing: An Application of Topological Data Analysis to Chronic Traumatic Encephalopathy

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# Contents

**1 Introduction** **2**
1.1 Chronic Traumatic Encephalopathy . . . 2
1.2 Topological Data Analysis . . . 2
1.3 Executive Summary . . . 4

**2 Data Processing** **5**
2.1 Extraction . . . 5
2.2 Pre-Processing . . . 5
2.3 Processing . . . 6

**3 Topological Data Analysis** **7**
3.1 Filtrations . . . 7
3.2 Barcodes and Diagrams . . . 7
3.3 Comparison . . . 8

**4 Discussion** **11**
4.1 Interdisciplinary Interpretation . . . 11
4.2 Limitations . . . 12
4.3 Reflection . . . 13

**Appendix A Mathematics of TDA** **14**
A.1 Simplicial Complexes . . . 14
A.2 Chain Complexes and Homology . . . 14
A.3 Persistent Homology . . . 16

**Bibliography** **17**

# Chapter 1

# Introduction

## 1.1 Chronic Traumatic Encephalopathy

Chronic Traumatic Encephalopathy (CTE) is a degenerative brain disease caused by repeated injury to the head. Sadly, it has been documented in athletes participating in contact sports, as well as military personnel, law enforcement, and victims of domestic violence. CTE has been noted as early as 1928, where it was termed *punch drunk syndrome*. [Mar28]

Early symptoms of CTE include short-term memory problems, executive dysfunction, depression, emotional instability, and suicidal behaviour. [SRD$^+$11, Table 3] Whilst CTE may be diagnosed on the basis of symptoms and a history of head injury, it is conclusively identified by gross and microscopic neuropathology upon autopsy.

Characteristics of the gross neuropathology of a brain with CTE include a decrease in brain mass, enlarged ventricles, and atrophy of particular parts of the brain. [SRD$^+$11, Table 1]

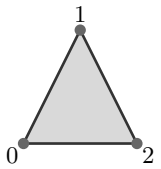
## 1.2 Topological Data Analysis

In this section, we present a brief introduction to topological data analysis (TDA). A comprehensive introduction to topological data analysis is given in [Nan]. TDA is a data analysis methodology that leverages techniques from algebraic topology to extract features related to the shape of a dataset. In this section, we sketch key concepts to motivate our subsequent analysis.

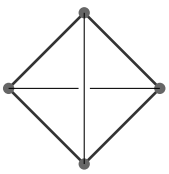
### 1.2.1 Simplicial Complexes

Topology is the study of the *shape* of mathematical objects. Formally, we require building blocks known as *simplicial complexes* (Definition A.1). Each simplicial complex (or simplex) has a dimension, and these form our understanding of shape.

- A 0 dimensional simplex is a point $\cdot$.
- A 1 dimensional simplex is a line $---$.
- A 2 dimensional simplex is a triangle




- A 3 dimensional simplex is a tetrahedron



TDA converts a smooth object, in this project a photograph of a brain, into a collection of simplicial complexes and studies their shape.

Morally, studying the shape of objects via topology is really about studying the *invariants* of objects. An invariant is a variable of an object that does not change when the object is deformed.

One invariant is the *Euler characteristic* (Definition A.3), which is a function from the set of simplicial complexes to the integers $\chi : S_{\text{spcx}} \to \mathbb{Z}$ given by:

$$\chi(X) = \sum_{i \geq 0} (-1)^i \ |X_i|,$$

where $|X_i|$ is the number of simplices of dimension $n$ in $X$. For example, the tetrahedron has one 3-dimensional simplex (the tetrahedron itself), four 2-dimensional simplices (the triangle faces), six 1-dimensional simplices (edges), and four 0-dimensional simplices (vertices). Therefore:

$$\chi(\Delta^3) = 4 - 6 + 4 - 1 = 1.$$

If one deforms the tetrahedron, within reason, the value of the Euler characteristic will always be 1. Thus, if one observes two objects that both have Euler characteristic 1, one can infer some topological similarity between them.

### 1.2.2 Chain Complexes and Homology

The invariants leveraged in topological data analysis come from homological algebra. The fundamental objects studied in homological algebra are chain complexes (Definition A.4). They take a non-linear object and place it in the realm of linear algebra. A chain complex is commonly represented by:

$$\cdots \xleftarrow{d_0} A_0 \xleftarrow{d_1} A_1 \xleftarrow{d_2} A_2 \xleftarrow{d_2} A_3 \dots$$

where $A_i$ are Abelian groups or modules and $d_i$ are homomorphisms. The $n$-th homology group (Definition A.7) is:

$$H_n = \frac{\ker(d_n)}{\text{im}(d_{n+1})}.$$

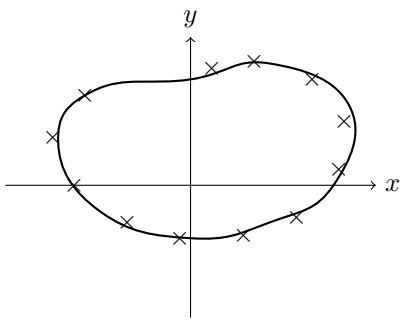
These homology groups, and their properties, are the invariants studied in TDA, for a certain choice of chain complex. A sequence is *exact* (Definition A.6) if every homology group is zero. The homology groups detect and classify the failure of exactness of the chain complex at each stage.

A fundamental example of homology is simplicial homology; see Definition A.9 and Example A.10.

### 1.2.3 Persistent Homology

Persistent homology is the core of TDA. Briefly, persistent homology is a scheme for building a graded chain complex from a point cloud, and then computing the appropriate homology groups from the chain complex. A detailed account of the development of persistent homology is given in [ZC05].

A point cloud is a sample of points $\times$ from a manifold. Visually:




The great miracle of persistent homology is that once one has computed the algebraic objects (Vietoris-Rips filtrations (Definition A.13) and persistence modules (Definition A.14)) there is a theorem (Theorem A.18) that allows the fundamental information from the persistence module to be expressed as a sum of its parts. Those parts are visually represented by a barcode. (Definition A.19) A schematic representation of this process is given in Fig. 1.1.

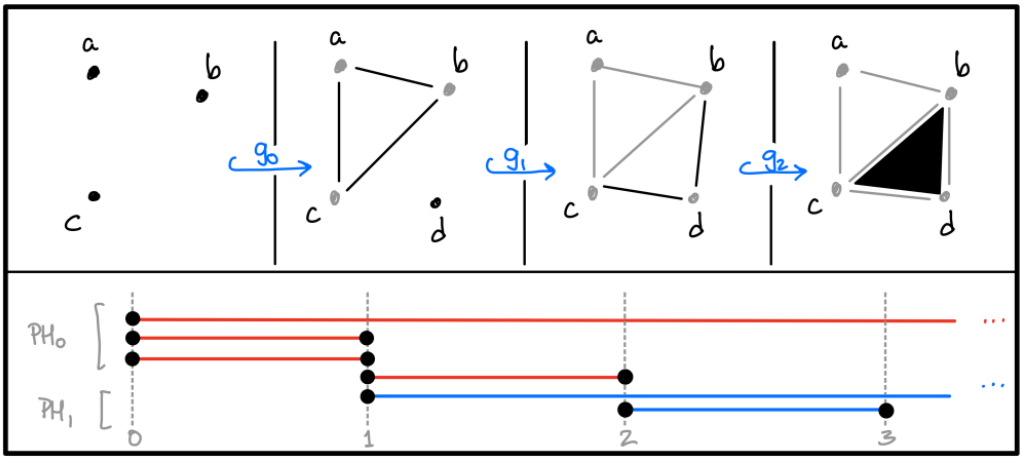



**Figure 1.1:** Schematic of persistent homology and barcodes. [Nan]

A collection of applications of topological data analysis to other domains is hosted at [GLR22]. Some articles that inspired this work are: an application of TDA to Alzheimer's disease detection [SYAS21], TDA as a means to interpret cytometry data in COVID-19 patients [MWDD22], and TDA as a means of identifying unique breast cancer subgroups. [NLC11]

## 1.3 Executive Summary

Data extraction and pre-processing is described in Section 2.1 and Section 2.2. Data processing and analysis was completed in a Jupyter notebook hosted in Google Colab running Python 3 with 12.7 GB of RAM.

This report analyses the coronal section of a brain from a patient with CTE and a control brain. Topological data analysis is implemented to convert the images to point clouds, convert the point cloud to a Vietoris-Rips filtration, compute the persistence module of the filtration, and present the decomposition of the persistence module via persistence barcodes and diagrams. The difference between the control and CTE barcodes/diagrams is measured by the bottleneck and Wasserstein distance. We interpret the significance of those distances by comparing them against distances from resampled control and CTE point clouds. We visualise the features that contribute greatly to the Wasserstein distance. Finally, we interpret the visualised features with the aid of clinician expertise.

# Chapter 2

# Data Processing

## 2.1 Extraction

The data for this report is an image of a coronal section of a normal brain and a brain with CTE, published by a team from the Boston University CTE Center and Brain Bank in [SRD+11, Fig. 1].

The brain with CTE is from a retired professional (American) football player, and whilst [SRD+11, Fig. 1] do not detail the patient of the control brain, we assume that it came from someone of the same sex and age at death as the CTE brain, as this is 'best practice' in the field.

We extracted the figure from the publication PDF file using an online tool. [Gee] The extracted JPEG file is in Fig. 2.1.

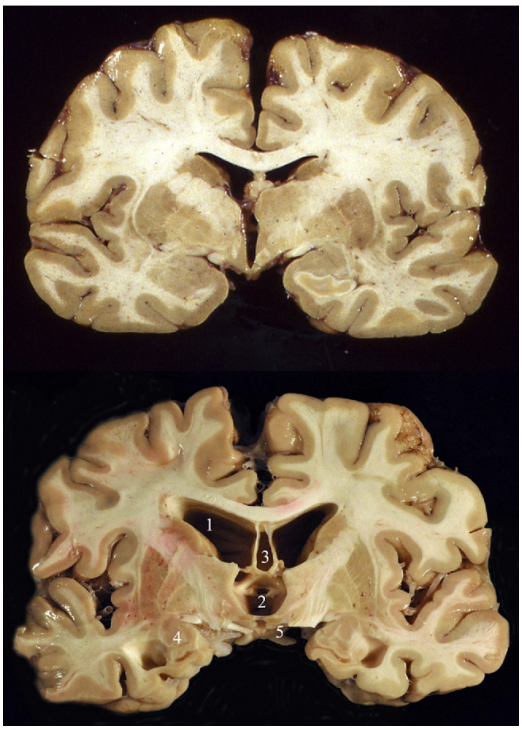



**Figure 2.1:** Extraction of [SRD+11, Fig. 1].

## 2.2 Pre-Processing

We performed two steps of pre-processing in Adobe Express Online. [Ado]

1. Fig. 2.1 was cropped into two separate images.
2. The authors of [SRD+11] annotated the CTE brain with numbers for use in-text. We used Adobe's *wand* tool to remove the annotations.

## 2.3 Processing

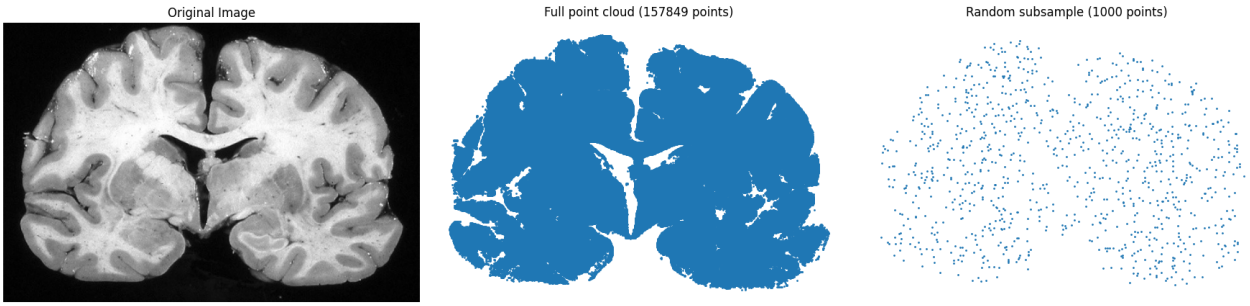
We processed the images in Python. The goal of processing was to convert the images from JPEG files to point clouds for use in TDA.

Firstly, we rendered the images in grayscale. This decision was made pragmatically, and the limitations are discussed in Section 4.2.

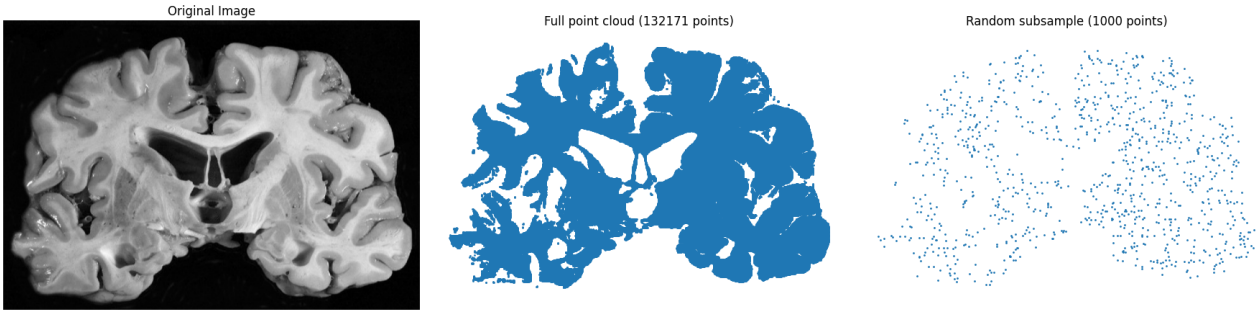
Secondly, we converted the grayscale image to a point cloud by a simple thresholding procedure. For each pixel in the image, a point was added to the point cloud if its brightness was greater than a certain value. We arbitrarily chose the threshold of 0.5, on a $[0, 1]$ scale. The limitations of this decision are discussed in Section 4.2.

Thirdly, we randomly sampled 1000 points of the point cloud. Sampling was necessary to keep computational time and RAM allocation under control, and 1000 points was the greatest number of points that did not crash the program.

These three steps, for the control and CTE brain, are presented in Fig. 2.2.




**(a)** Grayscale image, full point cloud, and sampled point cloud of the control brain.




**(b)** Grayscale image, full point cloud, and sampled point cloud of the CTE brain.

**Figure 2.2:** Visualisation of data processing steps.

# Chapter 3

# Topological Data Analysis

This chapter documents our topological data analysis of the two point clouds. Our main tool is the GUDHI library in Python. [The25] Each section is dedicated to a step of the analysis. Within each section, we

- Describe the computational steps employed by GUDHI,
- Explain the choice of any tuning parameters,
- Present results of the analytic step for our dataset.

## 3.1 Filtrations

The atoms of algebraic topology are chain complexes (Definition A.4), essentially sequences of algebraic information. In our analysis, we use the Vietoris-Rips complex (Definition A.12). The Vietoris-Rips complex packages the information from a point cloud in a format that is computable.

A filtration (Definition A.11) is a nested collection of algebraic objects, in our case Vietoris-Rips complexes. Thus, the goal of this step is to compute the Vietoris-Rips filtration (Definition A.13) following the GUDHI Tutorial: [CM25].

There are two tuning parameters in this process: *maximum edge length* and *maximum dimension*.

The maximum edge length, measured as a percentage of the diameter of the point cloud, represents the largest Vietoris-Rips complex included in the filtration. We selected the largest maximum edge length that did not crash the program, which was 15%.

The maximum dimension is the dimension of Vietoris-Rips complexes included in the filtration. We set a maximum dimension of 3, as dimension 0, 1, and 2 features are usually the only ones that are human-interpretable.

The filtration for both the control and CTE images use all 1000 points from the point cloud. The control filtration contained 24 million simplices and the CTE filtration contained 33 million simplices.

## 3.2 Barcodes and Diagrams

Persistence modules (Definition A.14) convert the information of filtrations to a format compatible with the tools of linear algebra. This has two significant consequences:

- The information stored in a filtration can be computed efficiently.
- That information can be decomposed as a sum of its parts via the structure theorem.

The decomposition of a persistence module is represented by persistence barcodes and diagrams. The two are different visualisations of the same information, and we followed the GUDHI Tutorial: [Pro25].

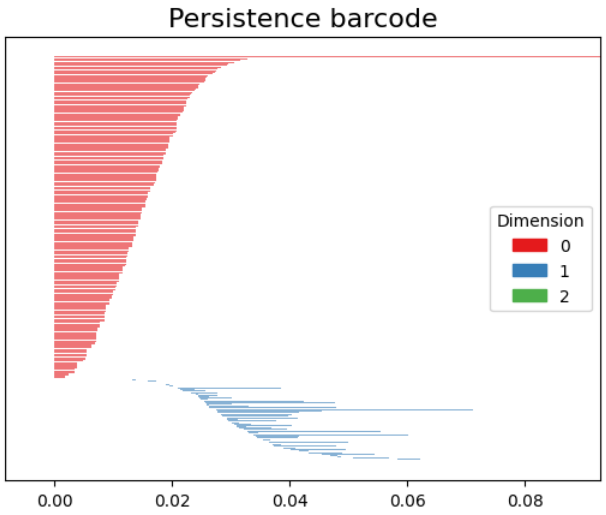
Computing the persistence module is the most expensive step of TDA. Prior decisions (for instance, how many points of a point cloud to subsample) were made with the persistence modules in mind.

The output of a persistence module computation is a list of features with the following information: birth

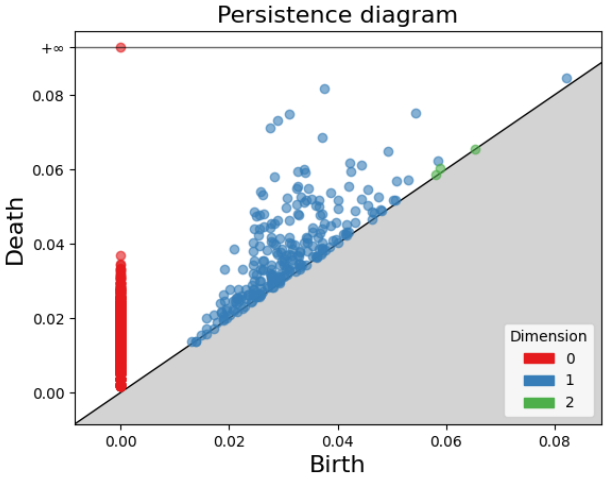
time, death time, and dimension. The birth time is the step at which that feature first appeared. The death time is the step at which that feature was filled in by a higher-dimensional feature. Features that do not die are assigned a death time of $+\infty$.

Persistence barcodes are a histogram of every feature along the vertical axis. The horizontal axis colours the feature from its birth time to its death time. Persistence diagrams plot the birth time of each feature along the $x$ axis and the death time of each feature along the $y$ axis. In both, the dimension of the feature is colour-coded.

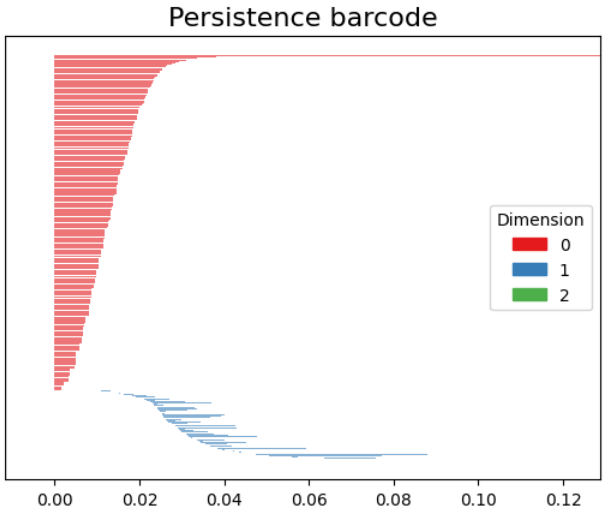
The persistence barcodes and diagrams of both the control and CTE brains are presented in Fig. 3.1.



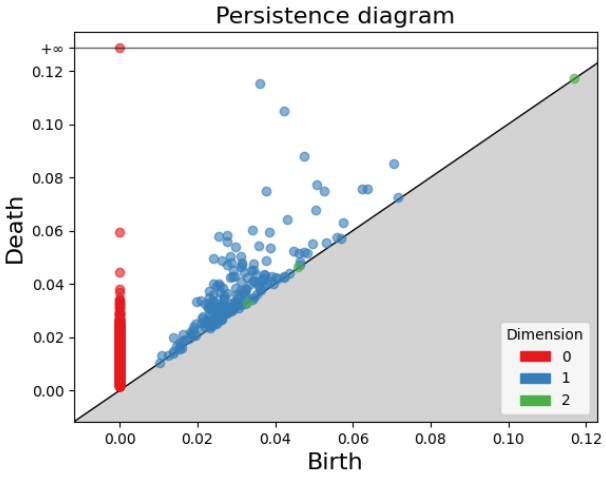

**(a)** Persistence barcode of the control brain.




**(b)** Persistence diagram of the control brain.




**(c)** Persistence barcode of the CTE brain.




**(d)** Persistence diagram of the CTE brain.

**Figure 3.1:** Persistence barcodes and diagrams of our data.

## 3.3 Comparison

Our comparison of persistence modules involves two questions: (i) are there any differences between the two brains? and (ii) can we locate and visualise those differences?

### 3.3.1 Are there any differences?

We need to compare diagrams/barcodes, which are derived from the persistence modules. We select two metrics: the bottleneck distance (Definition A.23) and the Wasserstein distance (Definition A.24). Both distances

attempt to match elements of the barcodes/diagrams under comparison. In the bottleneck distance, only the single greatest mismatch is considered, whereas in the Wasserstein distance all mismatches are weighted and summed.

Both distances are measured as a percentage of the point cloud diameter. Unlike familiar statistical tests, the $t$-test, $\chi^2$ test, etc, the bottleneck and Wasserstein distances do not have an analogous notion of hypothesis testing. Thus, to make comparisons, we construct *resampled* versions of both the control and CTE point clouds by selecting a different subsample of point to the original.

Another nuance is that persistence modules contain multiple types of topological data, graded by dimension. Thus, comparisons must be made on a dimension-by-dimension basis.

We present comparisons of the control vs CTE brain, and the control vs (control) resample and the CTE vs (CTE) resample, for each dimension, for both the bottleneck and Wasserstein distances, in Table 3.1 and Table 3.2.

| | Control vs CTE | Control vs Resample | CTE vs Resample |
|---|---|---|---|
| Dimension 0 | 2.274 | 0.375 | 0.999 |
| Dimension 1 | 3.329 | 0.815 | 2.511 |
| Dimension 2 | 0.068 | 0.068 | 0.075 |

**Table 3.1:** Bottleneck Distance (% of point cloud diameter)

| | Control vs CTE | Control vs Resample | CTE vs Resample |
|---|---|---|---|
| Dimension 0 | 0.142 | 0.040 | 0.034 |
| Dimension 1 | 0.200 | 0.114 | 0.168 |
| Dimension 2 | 0.052 | 0.034 | 0.057 |

**Table 3.2:** Wasserstein Distance (% of point cloud diameter)

We interpret the results of Table 3.1 and Table 3.2 dimension-by-dimension.

Dimension 0 features are born as individual points and die when they are connected with another point. In both Table 3.1 and Table 3.2 we see that the Control vs CTE distance is large relative to the resampled distances. Our interpretation of this is that many of the points in the CTE brain are isolated for 'longer', so there are greater distances between them.

Dimension 1 features are born as lines and die when they are filled in as a triangle. In Table 3.1 we see that the Control vs CTE distance is large relative to the resampled distances, whereas the difference is more modest in Table 3.2. This suggests that (relatively) few extreme features are driving the overall difference. These features are likely to represent the number and size of holes, as the triangles fill in the gaps created by the surrounding points.

Dimension 2 features are born as triangles and die when they are filled in as a tetrahedron. In both Table 3.1 and Table 3.2 we see that the Control vs CTE distance is similar to the resampled distances, so any observed dimension 2 features are likely just noise. Intuitively, this makes sense as the images are *flat* sections of a brain, which obscures any cavities.

### 3.3.2 Where are the differences?

Now our approach differs from many applications of TDA in the literature, where it is common to feed the features from TDA into a machine learning algorithm, for instance in [QTT+19].

We prefer to visualise the features of the point cloud that contribute most to the Wasserstein distance. We implement this procedure for the dimension 1 features, as they have the clearest interpretation.

To do this, we rank all dimension 1 features in the CTE brain by their contribution to the Wasserstein distance, highlight the dimension 1 simplex (line) that births the feature, and shade in the dimension 2 simplex (triangle) that kills the feature.

A histogram of the contributions to the Wasserstein distance of the dimension 1 features is plotted in Fig. 3.2.

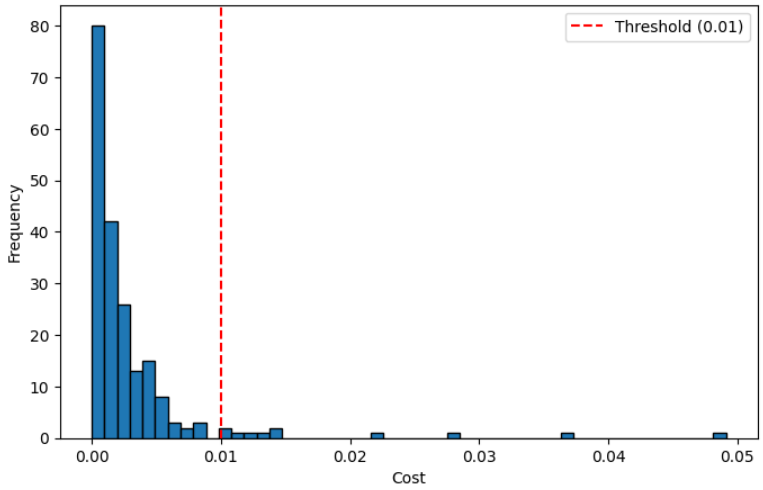



**Figure 3.2:** Histogram of Dimension 1 Features.

On visual inspection, we choose a threshold at 0.01 to represent the *greatest* contributors to the Wasserstein distance. This decision was also made to keep the number of features relatively small (11, in this case).

The birth dimension 1 simplex (line) and death dimension 2 simplex (triangle) corresponding to the 11 features above the threshold in Fig. 3.2 are plotted in Fig. 3.3.

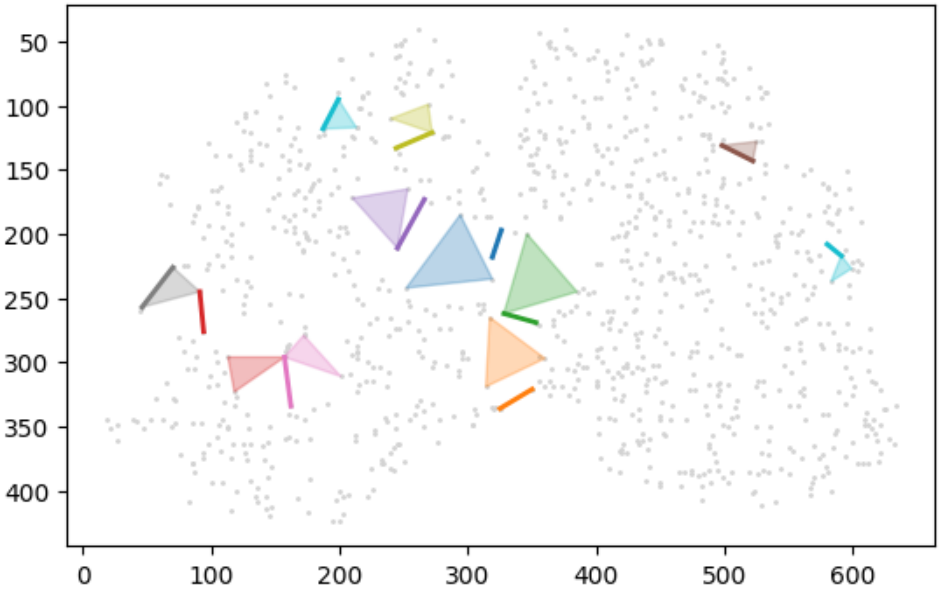


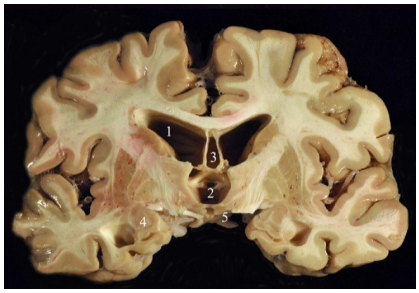

**Figure 3.3:** CTE Point Cloud with Highlighted Features.

# Chapter 4

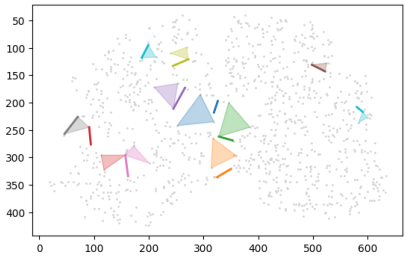
# Discussion

## 4.1 Interdisciplinary Interpretation

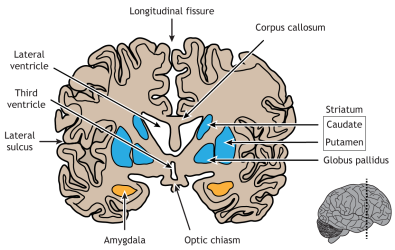
The section is based on a discussion with Dr. Jessica (Yi Xuan) Luo, psychiatry registrar at Northern Hospital, Epping. We compared the results of our analysis with [SRD+11, Fig. 1]. Fig. 4.1 collects the necessary information for this discussion.




**(a)** CTE brain from [SRD+11, Fig. 1]




**(b)** CTE Point Cloud with Highlighted Features




**(c)** Labelled coronal section of the brain [Hen24, Fig. 20.3]

**Figure 4.1:** Overall caption for the panel.

In Fig. 4.1a, the authors identify five differences between the gross pathology of the control and CTE brain:

1. Severe dilatation of ventricle II.
2. Severe dilatation of ventricle III.
3. Severe dilatation of cavum septum pellucidum.
4. Marked atrophy of the medial temporal lobe structures.
5. Shrinkage of the mammillary bodies.

Items 1,2, and 3 are characteristic of overall brain shrinkage. The ventricles and cavum septum pellucidum are responsible for nourishing the brain with cerebrospinal fluid. When that process is inhibited, overall brain function breaks down. In Fig. 4.1b, we observe features (blue, green, orange) that seem to coincide with Items 1, 2, and 3.

Item 4 describes the death of the temporal lobe. This includes the hippocampus, responsible for memory, and the amygdala, responsible for emotional regulation. Thus, it is not surprising that as that region is impacted those with CTE suffer from memory and emotional dysfunction. Whilst we observe features (light pink, dark pink) in Fig. 4.1b that are near this region, they seem to be focused on the dilatated lateral sulcus (see Fig. 4.1c), which is a corollary of overall brain shrinkage rather than a cause for specific concern. Atrophy refers to overall shrinkage, rather than creation of holes or cavities, which is why TDA may not have detected it.

Item 5 describes the shrinkage of the mammillary bodies, part of the limbic system, which manage many unconscious bodily instincts, like survival. Like Item 4, TDA does not detect this shrinkage.

Several other features (black, brown, light blue) are identified in Fig. 4.1b. However, each of them seem to be cavities created by overall brain shrinkage.

In this instance, TDA provided no advantage over a clinician looking at the images. TDA could not detect some features and detected others that were spurious.

## 4.2 Limitations

Two practical factors influenced many decisions in this report: (i) we had one semester (12 weeks) to complete the analysis and report, and (ii) computation time and power was limited.

Additionally, my learning goal for this report was to implement TDA in a practical setting. As such, some choices were made to maximise those goals, and in a professional setting other choices may have been better.

In this chapter, we discuss various limitations of this report.

1. Our study design compares only one CTE and one control brain. In part, this choice was made in the interests of time.

   There are also ethical concerns: many deaths due to CTE occur amongst high-profile athletes, and often their brains are donated to science on the condition of anonymity. Thus, outside of research institutes like Boston University's CTE Center or the University of Sydney's Australian Sports Brain Bank, large datasets are not available.

   Hypothetically, a similar analysis conducted on a larger dataset should also incorporate other variables, such as sport played, number of concussions, concussion symptoms, etc, into its analysis.

2. In Section 1.2, we introduced TDA as our main analytic tool for this report. TDA is not necessarily the best tool for this analysis, and there are several limitations worth discussing.

   - TDA renders a fundamentally smooth object (in this case, a brain) as a discrete mathematical object (a point cloud). There is no guarantee that the point cloud is a faithful approximation of the original smooth object. Mathematically speaking, a manifold or Riemannian surface may be a sounder representation, and analytic techniques under this assumption may be profitable, for example in [JWJ+25].
   - A central tenant of TDA is the use of topological, or shape-based, information. However, it is extremely difficult to interpret results in dimension $\geq 4$. In this report, we simply ignored these higher dimensional features, which potentially ignores important information.
   - TDA is inherently computationally expensive because the number of simplices formed from a point cloud grows exponentially. Thus, we really have only ourself to blame for many of the computational limitations.
   - By definition, TDA detects differences in shape that alter topological invariants, like the number of holes. Features that do not alter topological invariants, for example the overall shrinkage of a part of the brain, may therefore be missed by TDA.

3. In Section 2.2, we removed the author annotations on the CTE brain with Adobe Express Online's wand tool. We made this choice in the interests of time, but it introduces exterior noise to the image. An ideal approach would have been to contact the original authors of the paper and ask for a high resolution copy of the image.

4. In Section 2.3, we rendered the images in grayscale to simplify their conversion to a point cloud. Fundamentally, this ignores colour as a variable. An approach that may have mitigated this problem is *persistent homology for multispecies data*, a technique developed in [SDB+24], by treating different coloured pixels as different species of data.

5. In Section 2.3, we chose an arbitrary brightness threshold of 0.5. An alternative route could have been to repeat the analysis for a variety of thresholds and interpret the results as a collective.

6. In Section 3.3, we used a resampling procedure as a stand-in for parametric hypothesis testing. This could be improved by computing many resamples of the point clouds and reporting an empirical confidence interval from the collection of distances. We did not pursue this step due to the computational time required.

## 4.3 Reflection

In this report, I sought to extend my knowledge in three ways:

1. Topology is a field of mathematics that I am less familiar with. My Master's research was in algebraic combinatorics, so it was challenging to engage with a topic I did not have strong foundations in.

2. My Master's research was almost entirely theoretical. Whilst we used computer programming (mainly Mathematica) to check examples and build intuition, we did not have to make computational decisions or interpretations. Those goals, as well as coding in Python, was new to me in this report.

3. I chose an example from psychiatry in this report because it gave me an opportunity to engage in interdisciplinary research with my friend Dr. Jessica Luo. This process extended my professional and interpersonal skills, and improved the focus of the report.

The biggest challenge I faced in this report was bringing together the theoretical knowledge of TDA I had learnt with the practical considerations of being a beginner programmer with limited resources. Often, the theory told me what I *should* do, but I was limited in what I *could* do.

For example, I had learnt the theory of how to compare persistence diagrams but it was not clear how one should actually interpret the bottleneck or Wasserstein distance in practice. My solution was to resample the point clouds and use the Control vs Resample and CTE vs Resample distances as benchmarks for the Control vs CTE distances I was trying to understand.

# Appendix A

# Mathematics of TDA

In this appendix, we include mathematical definitions, theorems, and examples from lecture notes taken at the Australian Mathematical Sciences Institute Summer School at Monash University in January 2026. Dr Daniele Celoria and Dr Agnese Barbensi from the University of Queensland lectured on 'Computational and Combinatorial Algebraic Topology,' which introduced the mathematical theory underpinning TDA.

## A.1 Simplicial Complexes

**Definition A.1.** Let $n \in \mathbb{N}$. Define $[n] = \{0, 1, \dots, n-1\}$. A *simplicial complex* $X$ on $[n]$ is $X \subseteq \mathcal{P}([n])$ such that if $\sigma \in X$ and $\tau \leq \sigma$ then $\tau \in X$.

**Definition A.2.** The *dimension* of an $n-$simplex is $n$:

$$\dim \Delta^n = n.$$

The dimension of a simplicial complex is the dimension of the greatest $n-$simplex it contains.

$$\dim X = \max\{n \mid \Delta^n \subseteq X\}.$$

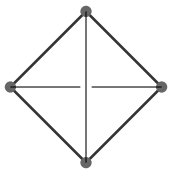
**Definition A.3.** The *Euler characteristic* is a function from the set of simplicial complexes to the integers $\chi : S_{\text{spcx}} \to \mathbb{Z}$ given by:

$$\chi(X) = \sum_{i \geq 0} (-1)^i \ |X_i|$$

where $|X_i|$ is the number of simplices of dimension $n$ in $X$. Moreover, the Euler characteristic is a specialisation of the *Poincare polynomial* $P : S_{\text{spcx}} \to \mathbb{Z}[t]$ at $t = -1$:

$$P(t) = \sum_{i \geq 0} |X_i| t^i.$$

For example, consider the $3-$simplex $\Delta^3$:



It has one 3-dimensional simplex (the tetrahedron itself), four 2-dimensional simplices (the triangle faces), six 1-dimensional simplices (edges), and four 0-dimensional simplices (vertices). Therefore:

$$\chi(\Delta^3) = 4 - 6 + 4 - 1 = 1.$$

## A.2 Chain Complexes and Homology

**Definition A.4.** A *chain complex* over $R$ (taken to be a ring with identity, usually $\mathbb{Z}, \mathbb{Q}, \mathbb{Z}/2\mathbb{Z}$) is a sequence of finitely generated $R-$modules $C_k, k \in \mathbb{Z}$ and morphisms $f_k : C_k \to C_{k-1}$ such that $f_k \circ f_{k+1} = 0$ for all $k$.

**Definition A.5.** Given an oriented simplex $\Delta^n = [v_0, \ldots, v_n]$, its *algebraic boundary operator* is:

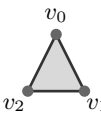
$$\partial \Delta^n = \sum_{i=0}^{n} (-1)^i [v_0, \ldots, \widehat{v_i}, \ldots, v_n]$$

where $\widehat{v_i}$ indicates that $v_i$ is removed, and the sum is treated as a purely formal sum.

For example,

$$\partial(\overset{v_0}{\circ} - \overset{v_1}{\circ}) = [v_0] - [v_1]$$

and

$$\partial\Big($$  $$\Big) = (\overset{v_1}{\circ} - \overset{v_2}{\circ}) + (\overset{v_0}{\circ} - \overset{v_1}{\circ}) - (\overset{v_2}{\circ} - \overset{v_0}{\circ}).$$



**Definition A.6.** If we have a sequence

$$\cdots \to C_{k+1} \overset{f_{k+1}}{\to} C_k \overset{f_k}{\to} C_{k-1} \ldots$$

we say it is exact in $k$ if

$$\ker f_k = \mathrm{im} f_{k+1}$$

which implies that $f_k \circ f_{k+1} = 0$.

**Definition A.7.** Let $(C, f)$ be a chain complex. The *homology group* of $(C, f)$ is the failure of exactness:

$$H_k(C, f) = \frac{\ker f_k}{\mathrm{im} f_{k+1}}.$$

Remark: $(C, f)$ exact in $k$ implies that $H_k(C, f) = 0$. Moreover, one commonly regards $\ker f_k = Z_k$ as cycles and $\mathrm{im} f_{k+1} = B_k$ as boundaries.

**Definition A.8.** If $X$ is a simplicial complex then

$$C_k(X) = \left\{ \sum_i a_i \sigma_i \mid a_i \in \mathbb{Z}, \dim \sigma_i = K \right\}$$

is the *simplicial chain group*, also known as the free group generated (over $\mathbb{Z}$) by the $K-$simplices. Moreover,

$$C(X) = \bigoplus_{k \in \mathbb{N}} C_k(X).$$

**Definition A.9.** The *simplicial homology* of $X$ is

$$H(C(X), \partial) = H_*(X; \mathbb{Z}),$$

where $*$ indicates that we may consider some $k$ or all of $k$.

**Example A.10.** Consider the simplicial complex $\Delta^1$. Its chain complex is given by:

$$\ldots 0 \to C_1(X) \to C_0(X) \to 0 \ldots$$

Which equates to:

$$\ldots 0 \to \mathbb{Z}\langle [v_0, v_1] \rangle \overset{\partial_1}{\to} \mathbb{Z}\langle [v_0], [v_1] \rangle \to 0 \ldots$$

Now, we can compute $\partial_1 : \mathbb{Z} \to \mathbb{Z}^2$ explicitly by

$$\partial_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

The kernel of $\partial_1$ is given by $\ker \partial_1 = 0$, so $H_1(\Delta^1; \mathbb{Z}) = 0$. Moreover, the image of $\partial_1$ is given by $\mathrm{im}(\partial_1) = -a[v_0] + a[v_1], a \in \mathbb{Z}$ so $\ker \partial_0 = \mathbb{Z}^2$. Thus, $H_0(\Delta^1; \mathbb{Z}) = \mathbb{Z}^2 / \mathrm{im} \partial_1 \cong \mathbb{Z}\langle [v_0] \rangle \cong \mathbb{Z}\langle [v_1] \rangle$. All in all,

$$H_*(\Delta^1, \mathbb{Z}) = \begin{cases} \mathbb{Z} & * = 0 \\ 0 & \text{otherwise} \end{cases}.$$

## A.3 Persistent Homology

**Definition A.11.** Let $(C, \partial)$ be a chain complex. A *filtration* on $C$ is:

$$\varnothing = F_0C \subseteq F_1C \subseteq \cdots \subseteq F_mC = C$$

such that $\partial(F_kC) \subseteq F_kC$, meaning that each $F_kC$ is a subchain complex. If $X$ is a simplicial complex then $F_kX$ is the *k-skeleton of X*.

**Definition A.12.** Let $(M, d)$ be a finite metric space. Then, the *Vietoris-Rips complex* $\mathrm{VR}_\epsilon(M)$ has vertices given by the set $\{X_0, X_1, \ldots, X_k\} \subset M$ and $k$-simplices are included if and only if $d(X_i, X_j) \leq \epsilon$ for all pairs.

**Definition A.13.** The Vietoris–Rips filtration is the nested family of simplicial complexes:

$$\{\mathrm{VR}_\epsilon(M)\}_{\epsilon \geq 0},$$

**Definition A.14.** An $\mathbb{N}$-indexed *persistence module* over a field $\mathbb{F}$ is a sequence $(V, a)$ of $\mathbb{F}$-vector spaces with linear maps between them:

$$V_0 \xrightarrow{a_0} V_1 \xrightarrow{a_1} V_2 \xrightarrow{a_2} V_3 \xrightarrow{a_3} \ldots$$

We say that a persistence module is of finite type if $\dim V_i < \infty$ and $a_i$ is an isomorphism for large enough $i$.

**Definition A.15.** For $i, j \in \mathbb{N}, j \geq i$ the $i, j$ persistent homology group of the persistent module $(V, a)$ is the subspace of $V_j$ given by:

$$\mathrm{PH}_{ij}(V, a) = \mathrm{im}(a_{ij}) \subset V_j,$$

where $a_{ij}$ is shorthand for the composition $a_{j-1} \circ a_{j-2} \circ \cdots \circ a_j : V_i \to V_j$.

**Definition A.16.** Several pieces of terminology are also commonplace:

- A cycle $v \in V_i$ is *born at filtration index i* if $v \notin \mathrm{im}(a_{i-1})$,
- A cycle $v \in V_i$ is said to *die at filtration index* $j \geq i$ if $j$ is the smallest index such that $a_{ij}(v) = 0$,
- If $a_{ij}(v) = 0$ never occurs then the *death time* of $v$ is $+\infty$,
- The *persistence* of $v$ is $j - i$.

**Definition A.17.** For all $0 \leq i \leq j \leq \infty$, $i \neq \infty$, the $\mathbb{N}$-indexed persistence module $(I^{i,j}, c^{i,j})$ given by:

$$I_p^{i,j} = \begin{cases} \mathbb{F} & i \leq p \leq j \\ 0 & \text{otherwise} \end{cases}$$

$$c_p^{i,j} = \begin{cases} \mathrm{id}_{\mathbb{F}} & i \leq p \leq j \\ 0 & \text{otherwise} \end{cases}$$

is called an interval module.

**Theorem A.18.** *For any finite type $\mathbb{N}$-indexed persistence module $(V, a)$ over $\mathbb{F}$, there exists a set of integer pairs $0 \leq i \leq j \leq \infty$ (with $i \neq \infty$) such that there is a unique (up to isomorphism of persistence modules) direct sum decomposition:*

$$(V, a) \cong \bigoplus_{[i,j]} (I^{i,j}, c^{i,j}).$$

*Proof.* See [Nan, Theorem 6.9]. □

**Definition A.19.** The collection of intervals $[i, j]$ (with multiplicity) generated by the Structure Theorem is the *barcode* of $(V, a)$.

**Definition A.20.** For $\epsilon \geq 0$, an $\epsilon$-interleaving between $(V, a)$ and $(W, b)$ consists of two families of functions:

$$\{\phi_t : V_t \to W_{t+\epsilon} \mid t \geq 0\}$$
$$\{\psi_t : W_t \to V_{t+\epsilon} \mid t \geq 0\}$$

with two requirements:

1. The parallelogram law:
   (a) For all $s \leq t$, $\phi_t \circ a_{s \leq t} = b_{s+\epsilon \leq t+\epsilon} \circ \phi_s$.

(b) For all $s \leq t$, $\psi_t \circ b_{s \leq t} = a_{s+\epsilon \leq t+\epsilon} \circ \psi_s$.

2. The triangle law:

   (a) For all $t$, $\psi_{t+\epsilon} \circ \phi_t = a_{t \leq t+2\epsilon}$.

   (b) For all $t$, $\phi_{t+\epsilon} \circ \psi_t = b_{t \leq t+2\epsilon}$.

**Definition A.21.** The *interleaving distance* $d((V,a),(W,b))$ is defined as the $\inf_\epsilon$ such that there exists an $\epsilon$-interleaving between $V$ and $W$.

**Definition A.22.** For $\epsilon \geq 0$ and $\epsilon$-matching between multisets $B, B'$ of intervals is $\rho : B_0 \to B_0^{'}$ a bijection with $B_0 \subset B$ and $B_0^{'} \subset B'$ such that:

- Every $[s,t]$ in $(B \setminus B_0) \cup (B' \setminus B_0^{'})$ has length $t - s \leq 2\epsilon$.
- If $\rho[s,t] = [s',t']$ then $|s - s'| \leq \epsilon$ and $|t - t'| \leq \epsilon$.

**Definition A.23.** The bottleneck distance $d_B(B,B')$ is the infimum over all $\epsilon \geq 0$ such that there exists an $\epsilon$-matching between $B$ and $B'$.

**Definition A.24.** Let $D, D'$ be persistence diagrams. Define the Wasserstein distance:

$$d_{WP}(D,D') = \min_{m \in M(D,D')} \left( \sum_{(a,b) \in M} ||a-b||_p^p + \sum ||s - \text{Proj}(s)||_p^p \right)^{1/p}.$$

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